



UK RESIDENTIAL SUSTAINABLE ENERGY CONSUMPTION MODELLING

*An econometric analysis of household gas and electricity
demand in the United Kingdom (1990-2030)*



Group E
David Jones - 38345676
Kori Lazar - 38869306
Oliver Heron - 39025659
Michal Kosnik - 38802139

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1. Executive Summary

This report presents an analysis of the UK's residential energy consumption to inform Government policy around the transition to sustainable energy. It addresses the issues around the 2050 Net Zero targets and the cost-of-living crisis, driven by over-reliance on imported gas. The objective is to analyse historical demand and forecast consumption from 2026 to 2030 informing government spending and focus.

Using a top-down approach, we created multiple regression models analysing gas and electricity demand. Our findings show that gas demand is highly inelastic, constrained by the UK's older, less insulated housing stock. Alternatively, electricity demand is more elastic and positively linked to rising disposable income. The model shows cross-price sensitivity: when gas prices rise, electricity demand rises. However, the UK's marginal pricing mechanism links retail electricity to wholesale gas prices, financially punishing consumers and limiting the transition to greener energy.

To address these issues, this report recommends providing specific government grants to overcome high upfront costs for home insulation and heat pumps, implementing reforms to decouple electricity from gas prices, and increasing public education on efficient appliance use.

2. Introduction

2.1 Background & Client Context

The United Kingdom's current energy landscape is characterised by issues between long-term environmental commitments and an ongoing cost-of-living crisis. While the nation has a mandate

to achieve Net Zero emissions by 2050 and agrees with the COP26 ambition of limiting global temperature increases to 1.5°C, UK households remain highly affected by extreme price volatility due to a reliance on imported international gas.

Specifically, the domestic sector is responsible for around 26.8% of the UK's greenhouse gas emissions, largely because 64% of homes still rely on natural gas boilers for heating. Framed as an advisory analysis for policymakers, this report asserts that accurate energy demand modelling is imperative. Such data-driven modelling is required to guide infrastructure investments and implement policies that accelerate decarbonisation while actively managing the financial burden on homeowners.

2.2 Rationale for Country Selection

The United Kingdom was selected as the location for this study because of three primary factors. First, there is a high availability of publicly accessible economic data provided by institutions such as the Office for National Statistics (ONS) and DESNZ. Secondly, the global gas price shocks experienced between 2021 and 2023 exposed problems within the UK energy system, showing there are deep rooted structural issues within the country. Finally, as residents and active stakeholders within the UK, we have firsthand awareness of the socioeconomic impacts being modelled, helping provide context when interpreting data.

2.3 Aims and Objectives

This project aims to analyse and forecast UK residential energy consumption using multiple regression models. The key objectives are to:

- Identify key demographic, economic, and physical factors influencing domestic gas and electricity consumption.
- Build a multiple linear regression model to analyse the impact of variables like household income and energy prices.
- Calculate expected residential energy consumption trends over the next five-year period (2026–2030).
- Deliver evidence-based policy recommendations to support the transition to sustainable energy and protect consumers from price shocks.

2.4 Project Scope and Boundaries

To maintain analytical focus, the system boundaries of this study are defined as:

- **In-Scope:** A top-down analysis of UK residential gas and electricity consumption using public secondary data. The methodology utilises multiple linear regression to evaluate socio-demographic, economic, and physical factors to generate a five-year forecast.

- **Out-of-Scope:** Energy consumption within the industrial, commercial, and transport sectors. This study also excludes primary household-level data collection and the modelling of short-term seasonal fluctuations.

3. Literature Review

3.1 Modelling Approaches (Top-Down vs. Bottom-Up)

This report by *Summerfield et al. (2009)* highlights the methodological differences between two modelling frameworks, 'bottom-up' and 'top-down'. While 'bottom-up' modelling approaches attempt to capture physical processes of building stocks by incorporating variables regarding building characteristics, 'top-down' approaches focus on core macro-drivers such as GDP and population growth. As demonstrated by *Summerfield et al. (2009)* through their research models, using a 'top-down' approach allows for superior transparency and interpretability, through the use of comparative macro variables such as external temperature and GDP. A key benefit of a 'top-down' approach is that models are based on actual consumption data, which is often easier to obtain and analyse, rather than theoretical building data, as seen in a 'bottom-up' model, which can be assumptive and difficult to obtain accurate data. Consequently, employing a 'top-down' approach provides a robust baseline that allows for research to take place and identify genuine consumption without being bounded by household variations. (*Summerfield et al. 2009*).

3.2 Review of Marginal Pricing

This publication by *Pikk & Viiding (2014)* explains that the UK's marginal pricing mechanism exposes the market to welfare detriments during global energy price shocks. It provides a foundational critique of the whole energy market design, which dissects the vulnerabilities associated with marginal costing within the market. The idea of marginal pricing links the retail price of electricity to the extreme volatility of global commodity markets, disproportionately exposing the consumer to global price shocks, even during periods where a large portion of the mix is made up of cheaper, renewable alternatives. Crucially, as the UK shifts towards a more renewable mix this pricing infrastructure becomes increasingly paradoxical, whereby there is more renewable energy produced, but is being charged to the market at the same price as more expensive gas production. *Pikk & Viiding (2014)* suggest that persisting with marginal costing in the long-run, as the system becomes more renewable dominant, will fail to provide any long-term price certainty which is required to incentivise capacity expansion, whilst failing to pass on economic benefits of a lower-carbon energy mix to the consumer. Consequently, any policy changes must consider decoupling electricity pricing from the traditional marginal pricing model, in order for long-term success, decarbonisation and price-certainty for the consumer (*Pikk & Viiding. 2014*).

3.3 Elasticity & Welfare Shocks

To enhance the predictive validity of a residential demand model, welfare and behavioural implications must be considered. Evidence from the report published by *Levell, O'Connell & Smith. (2025)* suggests that there is an average price elasticity of 0.31 indicating that energy demand is inelastic, with only a small change in demand due to fluctuating costs of energy. This focuses on behavioural adjustments in relation to income, where the study estimates, with no government intervention, that the average household faces an overall welfare loss of 6% of after-tax income, with the most vulnerable of households facing losses exceeding 11%. Notably, *Levell, O'Connell & Smith. (2025)* highlight how income alone is a poor indicator of these shocks, explaining only a small fraction of the variation in energy spending. For the UK energy consumption model, this necessitates the inclusion of variables beyond simple macroeconomic principles, such as housing efficiency which can be used to capture the heterogeneous nature of energy poverty. By integrating elasticity estimates, a proposed multiple regression model can be used to accurately simulate the trade-offs between price-driven demands and suppression, and the financial hardship that can threaten energy security. (*Levell, O'Connell & Smith, 2025*).

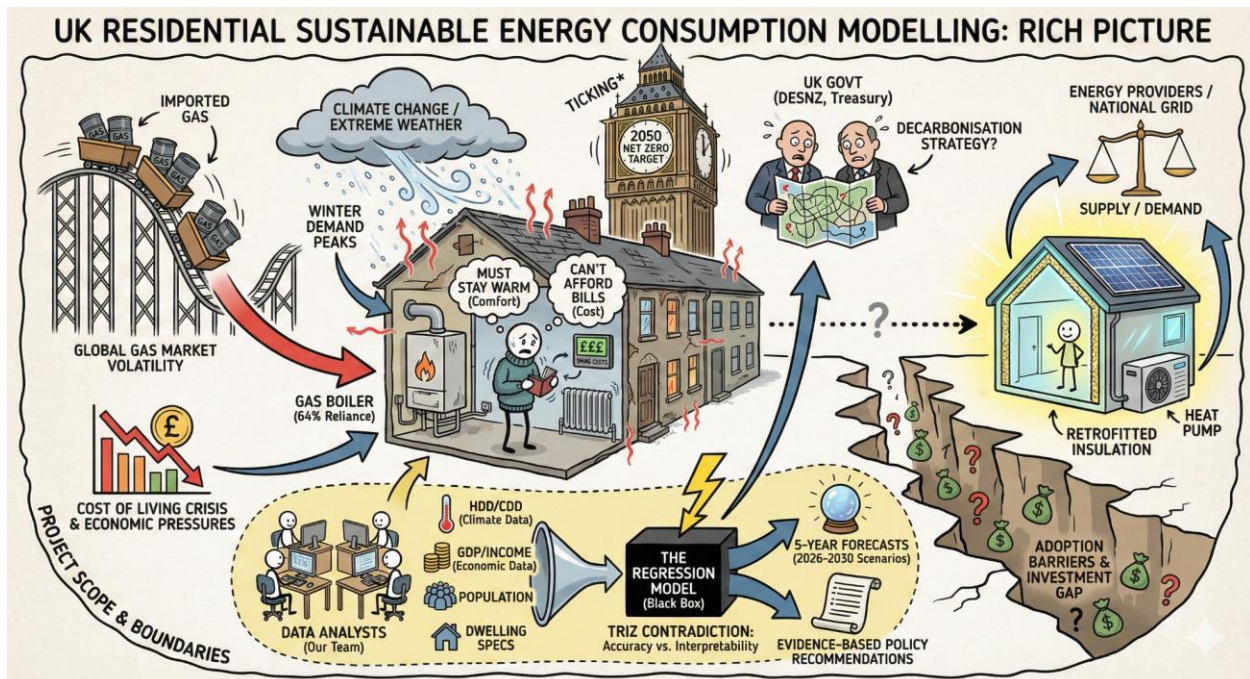
3.4 Gas Reliance & Old Housing Stock

The British energy market is characterised by a systematic vulnerability to external price shocks and long-run supply issues from global supply chain issues. According to the *Energy Crisis Commission (2024)*, the UK is 'dangerously underprepared' for future energy crises largely due to high over-dependency on volatile worldwide gas wholesale markets and aging housing stock which is ranked amongst the least efficient in developed Europe. The findings stated by the *Energy Crisis Commission (2024)* suggest that energy efficiency and price elasticity are increasingly dominant factors in the context of the UK energy market. The article highlights the previously mentioned issues with the aging housing stock, leading to households being forced into a reactive consumption cycle during periods of high cost, where consumers want to stay warm but are forced to cut heating during periods of high energy costs. This leads into the fact that any policy changes will have to be backed by an in-depth energy consumption model, and must go beyond simple market-based pricing, in order to recommend any valid policy interventions. (*UCL, Built Environment Faculty, 2024*).

4. Problem Structuring with Soft Systems Methodology

4.1 The 'Rich Picture'

As part of our Soft Systems Methodology to structure this complex problem, here is a Rich Picture which visualises the messy reality of the UK energy landscape, showing the perspectives of all the relevant stakeholders (*full-size image available in [Appendix 11.1](#)*).



(Figure 1)

It shows an outdated and overcomplicated system featuring old inefficient housing, conflicting consumer priorities, and government policies struggling to incentivise mass adoption of expensive, low-carbon technologies like heat pumps, solar panels and retrofitting old houses.

UK households are being directly hurt by volatility in energy prices, the gas market as well as climate and weather; they need to stay warm but are struggling to afford bills, and the barriers for adoption to retrofit more efficient technology are too high.

The government, energy providers and data analysts are working together to model this situation to forecast the most optimum solutions and provide evidence-based recommendations for policy makers. Meanwhile, the government is pushing for Net Zero and decarbonisation by 2050 which is regarded as an optimistic target, while the general population is trapped in a cost-of-living crisis partly due to the UK's reliance on volatile imported gas for heating.

4.2 CATWOE & Root Definition

To move from the messy reality of the Rich Picture to a structured system of interest, the CATWOE framework was applied. This explicitly defines the boundaries and purpose of the modelling system before conducting any data analysis:

- **Customers:** UK households (who benefit from stable prices) and policymakers (who benefit from data-driven insights).
- **Actors:** Data analysts (our project team) conducting the modelling.
- **Transformation:** Transforming inaccurate, fragmented or incomplete estimates into accurate, data-driven forecasts.
- **Weltanschauung / Worldview:** The belief that proactive, data-driven planning is essential to manage the transition to a sustainable energy future while protecting consumers from extreme price volatility.
- **Owner:** The UK Government / DESNZ, who own the ultimate policy goal of Net Zero.
- **Environmental Constraints:** The 2050 Net Zero legal targets, the reliance on existing inefficient housing stock, and extreme geopolitical energy price volatility.

Derived directly from this CATWOE analysis, the following root definition was created using the PQR formula (doing *What*, by *How*, for *Why*):

"A system owned by the UK Government and operated by data analysts, to transform inaccurate, fragmented or incomplete estimates of residential energy demand into practical, comprehensive data-driven energy forecasts using multiple linear regression analysis; in order to guide strategic policy interventions that protect UK households from price volatility while navigating Net Zero constraints."

4.3 Guiding the Analytical Focus

Breaking down the system, this way helped clarify the system of interest and guided our analytical focus, by helping us recognise that 'residential energy consumption' is not just one uniform behaviour. By mapping out the constraints, we realised that heating a home is a physical necessity generally tied to gas, whereas running modern appliances is a lifestyle choice tied to electricity and disposable income. Because of this, we knew a single 'total energy' model would fail. This insight directly dictated our methodology and analytical design: we had to split our data and build two entirely separate regression models.

5. Methodology

5.1 Data Collection & Sources

To get a reliable picture of UK domestic energy usage our study took a top-down approach, analysing national trends using publicly available data to achieve our goal of forecasting UK domestic consumption as a whole to create actionable policy recommendations. We built a dataset covering 35 years, from 1990 to 2024, to ensure the accuracy of our forecast and its reliability in identifying meaningful trends. This data set was made up of data from the following sources:

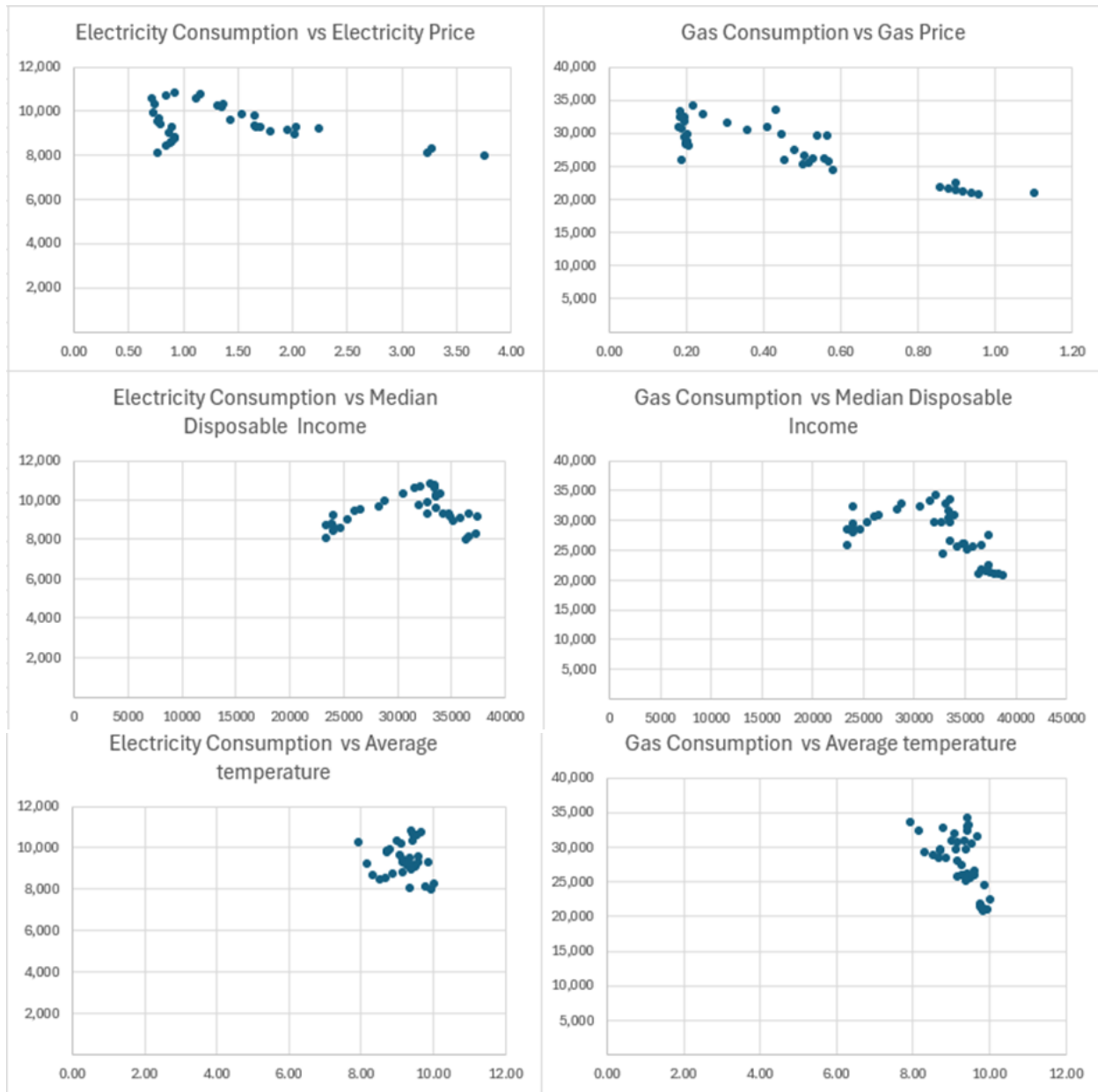
1. **DESNZ** - Overall consumption data and broken down into use case (*DESNZ, 2025a, 2025b*)
2. **ONS** - Economic factors such as Median disposable income, GDP, Population, etc. (*ONS, 2025*)
3. **Ofgem** - Historical Gas and Electricity Prices (*Ofgem, 2025*)

5.2 Analytical Approach

We decided to use an Ordinary Least Squares (OLS) multiple linear regression model for this project to uncover the factors that hold the most weight on the UK's domestic energy consumption, accurately forecast this consumption for the next 5 years and use this to provide data driven policy recommendations. We chose OLS because it can isolate the exact impact that movement in each individual factor can make on domestic consumption, this makes it the ideal tool to inform policy choices and predict policy outcomes across multiple scenarios. To further add to the usefulness of this model to the widest range of people, we had a large focus on making the output as intuitive as possible, ensuring all data was in its raw units (rather than converting to logarithms) allowing us to communicate our findings in real-world metrics such as gigawatt-hours rather than abstract percentages.

5.3 Exploratory Data Analysis (EDA)

Before making the multiple regression models we did an Exploratory Data Analysis (EDA) on the individual relationships between the independent predictors and the dependent variables (Gas and Electricity Consumption).



(Figure 2)

These scatter plots provided us with many insights:

- **Physical Drivers:** There is a clear, negative linear relationship between Average Temperature and Gas Consumption, which validates the assumption of domestic gas demand being in large part driven by physical heating requirements.
- **Price Elasticity:** Both Gas and Electricity consumption show negative trends when plotted against their respective prices. The data points are heavily clustered which is representative of volatile periods or price shocks over the 35 years analysed.
- **Income Effects:** The scatter plots for Median Disposable Income versus both Gas and Electricity consumption show disconnected clusters or inverted-U shapes instead of straight lines. We can see this because over the 35-years we are modelling appliances

became more efficient even as the population grew wealthier. Because these simple plots have this nuance it was clear that a simple linear regression would not be insufficient, and a multiple linear regression model would be needed to isolate the real impact of each predictor while holding the others constant.

5.4 Model Specifications

The selection process for what to be included in the model began with 15 potential macroeconomic and physical variables that we identified in part through the literature that we have previously discussed. A correlation analysis was done to filter out variables with weak relationships to energy consumption, from which 12 variables demonstrated strong correlations (above 0.8 or below -0.8). However, when running the regressions for gas and electricity with these 12 variables, clear signs of multicollinearity emerged in variables representing targeted use cases (e.g., natural gas for space heating, electricity for lighting) that were double counting the energy consumption data. This artificially inflated the model's explanatory power and impacted the accuracy of the coefficients.

To resolve this the model was fine-tuned through a backward stepwise elimination approach, removing the weakest and most insignificant predictors one by one. This process continued until all remaining variables achieved statistical significance (a p-value 0.05).

This left us with the same 5 variables for each model that proved to be statistically significant and were large drivers on the final consumption result. These were Electricity Price, Gas Price, the use of the *other* energy type (using gas use to predict electricity use, and vice versa), Median Income, and Average Temperature. The mathematical structures for the two models are defined as follows:

Electricity_Consumption

$$= \beta_0 + \beta_1(\text{Electricity_Price}) + \beta_2(\text{Gas_Price}) + \beta_3(\text{Gas_Consumption}) \\ + \beta_4(\text{Median_Income}) + \beta_5(\text{Average_Temperature}) + \epsilon$$

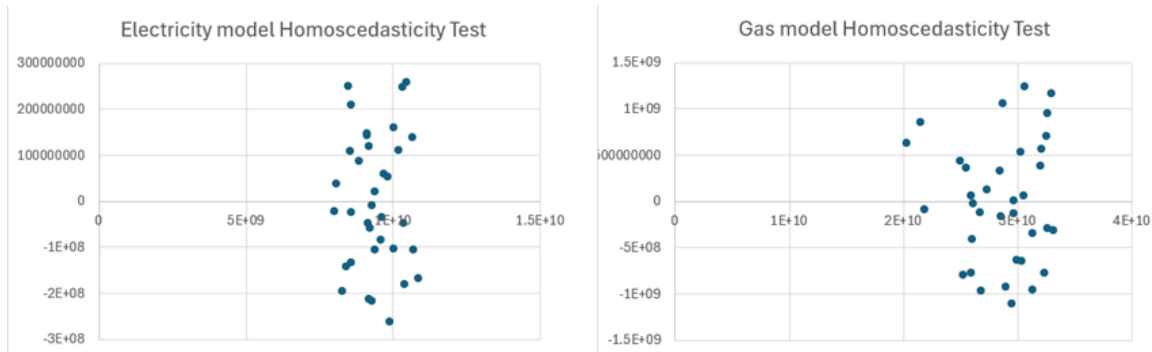
Gas_Consumption

$$= \beta_0 + \beta_1(\text{Gas_Price}) + \beta_2(\text{Electricity_Price}) + \beta_3(\text{Electricity_Consumption}) \\ + \beta_4(\text{Median_Income}) + \beta_5(\text{Average_Temperature}) + \epsilon$$

5.5 Testing Regression Assumptions

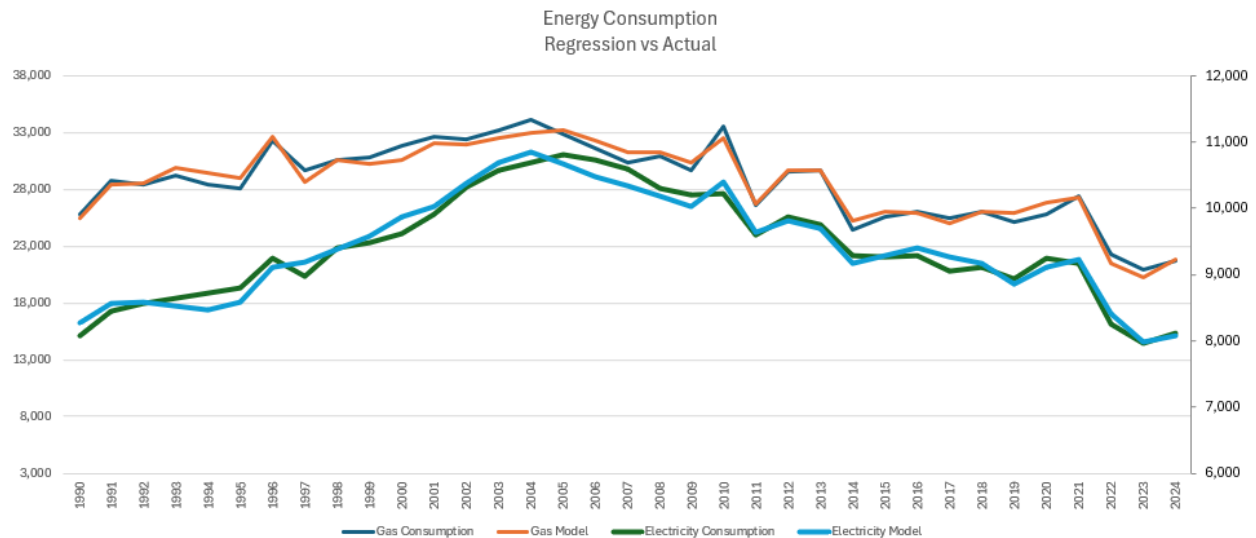
To ensure the statistical validity of both the electricity and gas models we evaluated both against core assumptions of multiple linear regression which confirmed that the final 5-variable models are suitable for policy forecasting:

- **Linearity:** In the Exploratory Data Analysis, the scatter plots confirmed a linear relationship between the dependent variables and the continuous independent predictors.
- **Homoscedasticity:** To test that the variance of the residuals remains constant across all levels of the independent variables, a Residuals vs. Fitted Values plot was inspected. The plot displayed a random, even scatter of points around the horizontal axis with no distinct funnel shapes, confirming homoscedasticity.



(Figure 4)

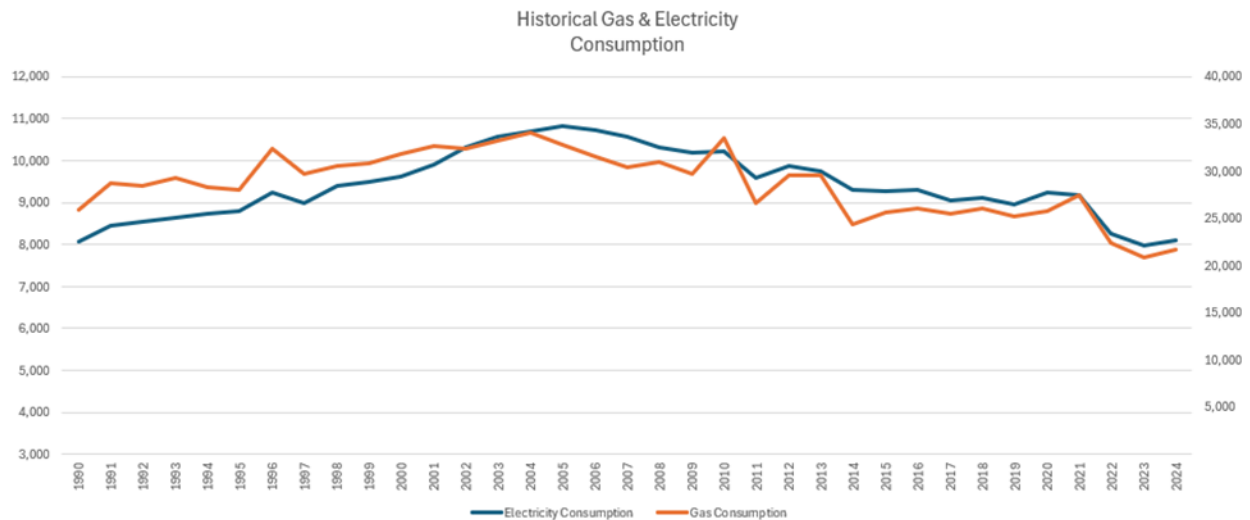
After running our final models, we found based on the R-Square that they could explain 96% and 97% of the movements in energy consumption respectively. To prove its accuracy in this explanation we tested the error rate, comparing the past energy consumption data against predictions of those same years produced by the model's coefficients. From this we found the error rate across both models did not exceed 3% for all 35 years of historical data, the actuals and regression outputs are plotted below.



(Figure 5)

6. Findings

6.1 Historical Trends / Context



(Figure 6)

Analysing the historical data spanning from 1990 to 2024 reveals the necessity for an inelastic nature of domestic energy consumption in the UK but also the severe volatility of the energy market due to factors outside of economic drivers such as geopolitics. Domestic energy consumption demonstrates a steady downward trend, driven by gradual efficiency improvements in appliances and rising baseline cost. However, we can see the impact geopolitical instability can have on domestic energy consumption between 2021 and 2023 where a surge in the energy price led to a large drop in consumption, exacerbated by the UK's existing cost of living crisis. Despite these significant price shocks total domestic energy demand in the UK did not collapse proportionally, highlighting that heating remains a fundamental inelastic necessity, largely because approximately 64% of UK households rely on aging gas boilers, leaving a vast majority of the population structurally vulnerable to the volatility of global wholesale gas markets.

6.2 Electricity Regression Output

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.983635674							
R Square	0.967539139							
Adjusted R Square	0.961942439							
Standard Error	158247756							
Observations	35							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	5	2.16462E+19	4.32924E+18	172.8767155	1.16622E-20			
Residual	29	7.26228E+17	2.50424E+16					
Total	34	2.23724E+19						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	-2283472672	908147790.1	-2.514428485	0.02	-4140843452	-426101892.5	-4140843452	-426101892.5
Electricity Price	-1001567434	139067829.3	-7.202006666	0.00	-1285993081	-717141787	-1285993081	-717141787
Gas Price	2333709351	457727595.6	5.098467677	0.00	1397551304	3269867397	1397551304	3269867397
total natural gas consumption	0.200010483	0.017203416	11.62620727	0.00	0.164825546	0.23519542	0.164825546	0.23519542
Median Disposable Income	105867.8089	11588.48101	9.13560706	0.00	82166.70404	129568.9138	82166.70404	129568.9138
Average temperature	344155676.4	75470602.31	4.560128923	0.00	189800963.5	498510389.4	189800963.5	498510389.4

(Figure 7)

The regression analysis for domestic electricity consumption had an Adjusted R-Square of 0.96 meaning its variables can explain 96% of the variance in consumption. The output identified price as the primary mover of demand and allowed us to calculate that for every £1 per unit increase in electricity costs, domestic demand falls by approximately 100 Gigawatt-hours, an amount roughly equivalent to the annual consumption of 37,000 homes. Through digging into this further the model uncovered a substitution effect between the two energy commodities; a £1 increase in gas prices results in a rise of 233 Gigawatt-hours (equivalent to 81,000 homes) in electricity demand. This indicates that as gas becomes unaffordable, consumers actively shift their consumption behaviours toward secondary electrical sources, such as plug-in space heaters.

6.3 Gas Regression Output

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.98150075							
R Square	0.96334372							
Adjusted R Square	0.95702367							
Standard Error	717922706							
Observations	35							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	5	3.92813E+20	7.85627E+19	152.4266404	6.75413E-20			
Residual	29	1.4947E+19	5.15413E+17					
Total	34	4.0776E+20						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	1.6724E+10	3321293005	5.035238565	0.00	9930695723	23516309533	9930695723	23516309533
Gas Price	-1.07E+10	2056598870	-5.202672388	0.00	-14906027127	-6493593185	-14906027127	-6493593185
Electricity Price	3592846548	815391198.7	4.406285662	0.00	1925184299	5260508798	1925184299	5260508798
total electricity consumption (Gtoe)	4.11654641	0.354074748	11.62620727	0.00	3.392382239	4.84071058	3.392382239	4.84071058
Median Disposable Income	-362747.974	78619.12864	-4.61399128	0.00	-523542.1464	-201953.802	-523542.1464	-201953.802
Average temperature	-1781094921	303150343.3	-5.875285845	0.00	-2401106989	-1161082853	-2401106989	-1161082853

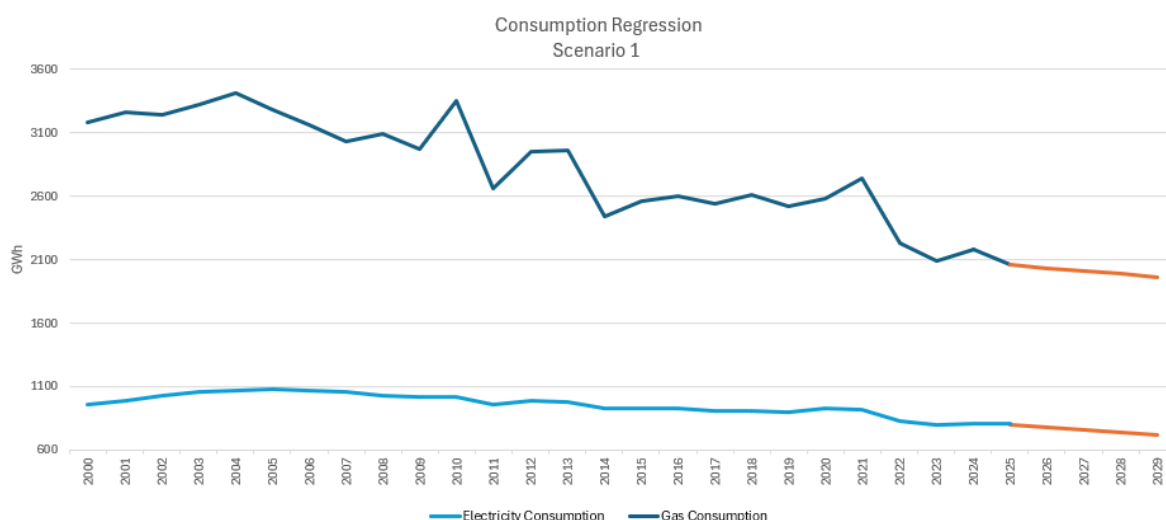
(Figure 8)

Similarly, the regression analysis for domestic gas consumption had an Adjusted R-Square of 0.96 meaning its variables can explain 96% of the variance in consumption. From this model we found a clear "*Wealth Effect*", through median disposable income having a negative coefficient which suggests that as UK households grow wealthier, they naturally tend to move away from the consumption of gas toward modern, electric alternatives. However, the regression also highlights a risk towards decarbonization targets: the interaction between the prices of Gas and Electricity on each other's consumption means high electricity prices can inadvertently drive consumers back toward established gas infrastructure despite the willingness to reduce gas consumption. Therefore, if electricity remains too expensive, households are financially disincentivised from adopting heat pumps, which threatens to stall the national Net Zero agenda.

6.4 5-Year Forecast (2025-2030)

We used the above-mentioned regression models to forecast energy demand from 2024 to 2029 for Electricity and Gas under two primary scenarios.

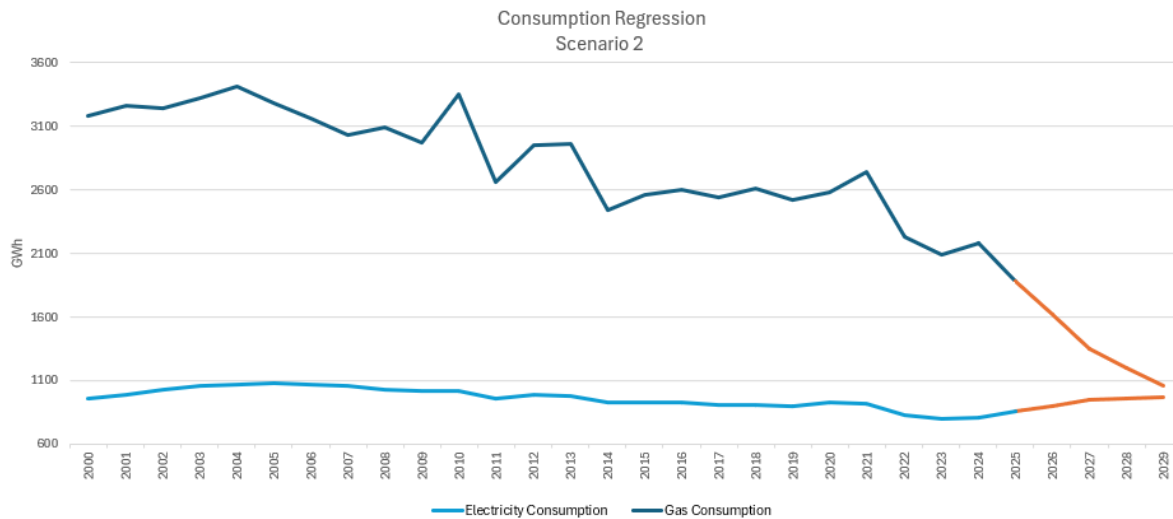
- Scenario 1 (Status Quo):** Without policy intervention and assuming electricity prices continue to increase the model predicts that heavy reliance on gas will persist. In this scenario the forecast predicts a 10% decrease in gas consumption and an 11% decrease in electricity consumption between 2024 and 2029. The decline in consumption across electricity and gas reflects the shrinking of energy demand as due to economic factors such as the cost-of-living crisis, rather than a successful green transition as consumers are forced to use less energy overall to make ends meet. The UK population lacks the disposable income required to overcome the "*green premium*" associated with transitioning to sustainable infrastructure like heat pumps or electric vehicles.



(Figure 9)

- Scenario 2 (Policy Shift):** However, if targeted policy changes get electricity prices back down to pre-Covid levels and the government proactively subsidises necessary

infrastructure (such as EV charging networks and residential heat pump grants), the forecast predicts a dramatic consumption shift. In this scenario, the forecast shows a 51% reduction in gas consumption with a 20% increase in electricity consumption by 2029, showing what a market-driven approach to reduce our reliance on fossil fuels could look like. As electricity becomes the more economical energy source households will actively substitute gas for electricity, acting as a cheaper and safer option as the UK consumer is less impacted by future geopolitical gas price shocks.



(Figure 10)

7. Discussion

7.1 Interpretation of Results

The regression model highlights a distinct divergence in domestic energy trends. Electricity demand shows a positive correlation with median disposable income: as consumers grow wealthier, increased acquisition of electronics and modern appliances is inherently associated with higher baseline electricity usage.

Conversely, gas consumption operates primarily as a highly inelastic physical necessity. This is tied to the UK's ageing, poorly insulated housing stock, which requires a high level of British Thermal Units (BTUs) to maintain comfortable temperatures, leaving households reliant on burning gas regardless of price. However, a negative coefficient for disposable income regarding gas consumption implies that wealthier populations naturally make the transition away from gas reliance, an association likely linked to investments in retrofitting, insulation, or electric heat pumps.

Crucially, the model reveals a cross-price sensitivity: for every £1 increase in gas prices, electricity demand rises by approximately 233 Gigawatt-hours. This substitution effect suggests households

resort to electric plug-in heaters when central gas heating becomes unaffordable. Within the UK's marginal pricing mechanism, where wholesale electricity is often pegged to expensive gas generation, global gas volatility strongly correlates with inflated electricity bills. If these inflated electricity prices persist, they risk stalling decarbonisation by removing the financial incentive to abandon established gas infrastructure.

To investigate this substitution effect, the model intentionally included the opposing energy type's total consumption as an independent variable. This choice tries to capture interconnected household behaviours, such as replacing gas central heating with supplementary electrical heating, that broad price indices might miss, albeit introducing some statistical constraints.

7.2 Comparison with Existing Literature

The model's indication of highly inelastic gas consumption aligns strongly with the findings of the Energy Crisis Commission (*UCL, Built Environment Faculty, 2024*), which characterises the UK as highly vulnerable due to its ageing, inefficient housing stock. This structural inefficiency traps households in what the literature describes as a "*reactive consumption cycle*." This agrees with our model's finding that gas usage is strongly linked to physical necessity and thermal constraints rather than purely choice. Furthermore, the inelastic nature of this demand is supported by *Levell, O'Connell, and Smith (2025)*, who estimate an average price elasticity of just 0.31. Our model's outputs agree with their assertion that such inelasticity exposes households to severe welfare losses during price shocks, as consumers cannot easily reduce their baseline heating requirements.

Regarding socio-demographic factors, the finding that electricity demand rises with disposable income adds nuance to the assertions of *Levell, O'Connell, and Smith (2025)*. They note that income alone is often a poor predictor of total energy spending variation. Our model clarifies this dynamic by disaggregating the fuels: wealth shows a positive association with baseline electricity consumption, likely via lifestyle choices and appliance adoption, but does not inherently correlate with gas consumption, highlighting the heterogeneous nature of domestic energy poverty.

Crucially, the model's observed cross-price sensitivity, where electricity demand rises alongside gas prices, illustrates the vulnerabilities of the UK's marginal pricing mechanism outlined by *Pikk and Viiding (2014)*. By linking retail electricity to volatile global commodity markets, the current system disproportionately exposes consumers to external shocks. Our analysis indicates that without the pricing decoupling recommended by Pikk and Viiding, the economic benefits of a renewable energy transition may fail to reach consumers, potentially stalling broader decarbonisation efforts.

Finally, the successful identification of these macroeconomic trends helps validate the methodological framework endorsed by *Summerfield et al. (2009)*. The top-down approach provided a strong, baseline of aggregate consumption, avoiding the complexities associated with bottom-up models and information.

7.3 Implementation Awareness & Project Adaptation

During the progression of this study, several methodological and data-related challenges emerged that required changes. Firstly, the identification of multicollinearity during the initial modelling phase. When incorporating specific consumption variables alongside broader economic indicators, the model exhibited overlapping patterns, which threatened the reliability of the outputs. To resolve this and ensure statistical robustness, we iteratively adapted the model, filtering the dataset down from over 16 initial factors to the 5 most statistically significant variables.

Furthermore, sourcing consistent, long-term historical data for specific independent variables presented a distinct challenge. For instance, where long-term household price data was fragmented or unavailable, we had to adapt by calculating implied prices to ensure the regression analysis could proceed. While this data engineering allowed for the creation of a functional model, it introduces a reliance on manually calculated inputs. Navigating these constraints was essential to maintaining the structural integrity of the baseline 5-year forecast.

8. Assumptions & Limitations

8.1 Methodological Limitations

Firstly, one limitation of this study is the exclusion of Heating Degree Days (HDD). By relying on average temperature, the model assumes a relatively flat weather baseline, failing to capture the extreme winter demand peaks that can stress the energy grid and household finances.

Furthermore, the top-down approach introduces aggregation bias. By utilising national-level data, the model cannot account for small scale variations or the impact of local competition, such as disruptive energy providers altering localised consumer behaviour. This reliance on macro-data also risks omitted variable's including relevant qualitative factors, (like pro-environmental attitudes) which are unquantifiable and therefore excluded. Additionally, reporting discrepancies between rural and urban areas may subtly skew the national averages.

Another limitation of the regression specifications is the inclusion of gas consumption as a predictor in the electricity model, and vice versa. While it was intended to capture behavioural substitution, both variables are fundamentally driven by the exact same factors (e.g., temperature, income, and housing inefficiency). Consequently, the analysis captures shared seasonal and economic patterns rather than the isolated effect of one fuel upon the other. This means the exceptionally high Adjusted R-Squared values (approx. 0.96) could be artificially inflated by this shared variance.

Finally, OLS regression modelling is inherently linear. Real-world energy transitions, such as the mass adoption of heat pumps or the effects of policy interventions, frequently exhibit non-linear trajectories. The model's structural simplicity may therefore struggle to precisely project the long-term impacts of green infrastructure integration.

8.2 Data Limitations

A notable limitation regarding data integrity stems from the necessity to manually calculate certain regression inputs, such as historical energy prices. While the analysis evaluates 35 years of historical data, calculating implied prices rather than utilising explicit, pre-existing datasets inherently introduces potential error into the model's foundation.

Additionally, the historical datasets contain a significant weakness due to the COVID-19 pandemic. The sudden transition to working from home in 2020 fundamentally changed domestic consumption patterns across the UK. This widespread behavioural shift introduces major volatility into the historical trendline, which may distort the baseline averages used in future projections.

Finally, the purely top-down approach restricts the model's ability to analyse the effects of small-scale changes. Sudden technological or regulatory shifts, such as the phased introduction of LED lighting standards, create household-level efficiency gains that are difficult to isolate or explicitly account for within utilising national data.

8.3 Forecasting Assumptions

The projections for the 2026-2030 period rely on the assumption that historical trends will continue without major external shocks. For instance, to generate the baseline forecast, the model assumes that variables such as median disposable income and population will grow linearly based on historical averages. In a volatile global economy, these linear assumptions may not hold true.

Crucially, the alternate forecast scenario, which models a decrease in electricity prices causing a shift away from gas, assumes a successful decoupling of electricity prices from gas prices. It is important to acknowledge that altering this marginal pricing mechanism is a highly complex political challenge, not merely an economic model adjustment. Many governments have attempted market reforms without long-term success, making this a significant real-world barrier to realising the forecasted demand shifts.

Furthermore, any projected transition away from gas assumes that the required physical infrastructure, such as localised grid capacity and EV charging's, will be deployed rapidly enough to meet rising electricity demand. If policy or infrastructure development lags, these forecasts will likely incorrectly estimate the speed of residential decarbonisation.

9. Conclusion, Recommendations, and Further Steps

9.1 Conclusion

This project has successfully delivered a robust multiple linear regression model and a five-year baseline forecast to evaluate the primary drivers of UK residential energy consumption. The analysis reveals a fundamental divergence in domestic energy use. Specifically, gas demand remains highly inelastic and is heavily constrained by the physical realities of an ageing housing stock, whereas electricity demand demonstrates greater elasticity and a positive correlation with growing disposable income. Crucially, the model highlights a major vulnerability stemming from the UK's marginal pricing mechanism. By pegging retail electricity to wholesale gas markets, the current framework creates a situation where high gas prices inadvertently inflate electricity costs. This dynamic financially penalises consumers and actively risks stalling the broader transition to Net Zero, demonstrating the urgent need for targeted, data-driven policy interventions.

9.2 Recommendations for Policymakers

The UK residential energy market is highly vulnerable to global price volatility due to its structural dependence on gas. Because marginal pricing ties the electricity price to the most expensive method of production, this can lessen the reduced pricing effects of renewable generation. As the amount of renewable generation increases, this can help to reduce the magnitude and frequency of price spikes. If energy prices were to return to pre-2021 levels, as assumed in Scenario 2 of our forecast, and the growth of renewable energy production continues at an expanding rate, it is feasible for renewables to consistently supply over 50% of the energy supply by 2029. In turn, this means the removal of marginal pricing can be looked at as a long-term reform as renewables continue to be an increasing part of the mix.

Efficiency improvements are another recommendation for reducing consumption and mitigating system stress. Interventions such as insulation upgrades, heat-pump instalment and improved efficiency of boilers will together combine to create a more efficient energy system. As noted in the findings, as electricity prices decrease this will encourage more households to switch their heating methods from gas to electric, therefore increasing efficiency and saving on bills. These effects should be focused on older, poorly insulated homes, often with low-income residents, however policy intervention is required to provide upfront investment for household changes.

Behavioural change forms the third key recommendation for reducing overall consumption and supporting a more resilient heating system. Adjustments in education such as improved management of heating schedules, reductions in hot water usage and wastage, and more efficient operations of household appliances, each contribute small reductions in demand, but when combined on a national level, the cumulative effect becomes significant, easing pressure on the

system during periods of high demand. Within the policy shift forecasting scenario, it is evident that all efficiency improvements are vital to prevent unnecessary energy growth, if small behavioural changes reduced demand by 3-5% this would offset some of the electricity growth, therefore allowing renewables to cover a larger share of overall consumption.

9.3 Further Steps for the Model

Future iterations of this residential energy model must incorporate a specific climate variable, such as Heating Degree Days (HDD). Relying solely on average temperatures limits the model's ability to account for severe seasonal demand volatility.

Furthermore, as the UK increases its renewable infrastructure to decrease reliance on gas, future models must also account for the weather-dependent nature of wind and solar generation on the supply side.

Transitioning from a purely top-down approach to a hybrid framework would also significantly enhance forecasting accuracy. By integrating bottom-up data, such as the National Grid's Future Energy Scenarios (FES), the model could better account for other limitations like the adoption rates of electric vehicles and residential heat pumps.

In addition, because gas and electricity demand influence each other simultaneously, future models should make use of advanced statistical techniques (Vector Autoregression) to separate the overlapping relationship, resulting in a more accurate and reliable baseline forecast.

Finally, introducing regional changes are recommended to investigate how geographic temperature differences and regional income divides affect localised energy usage.

10. References

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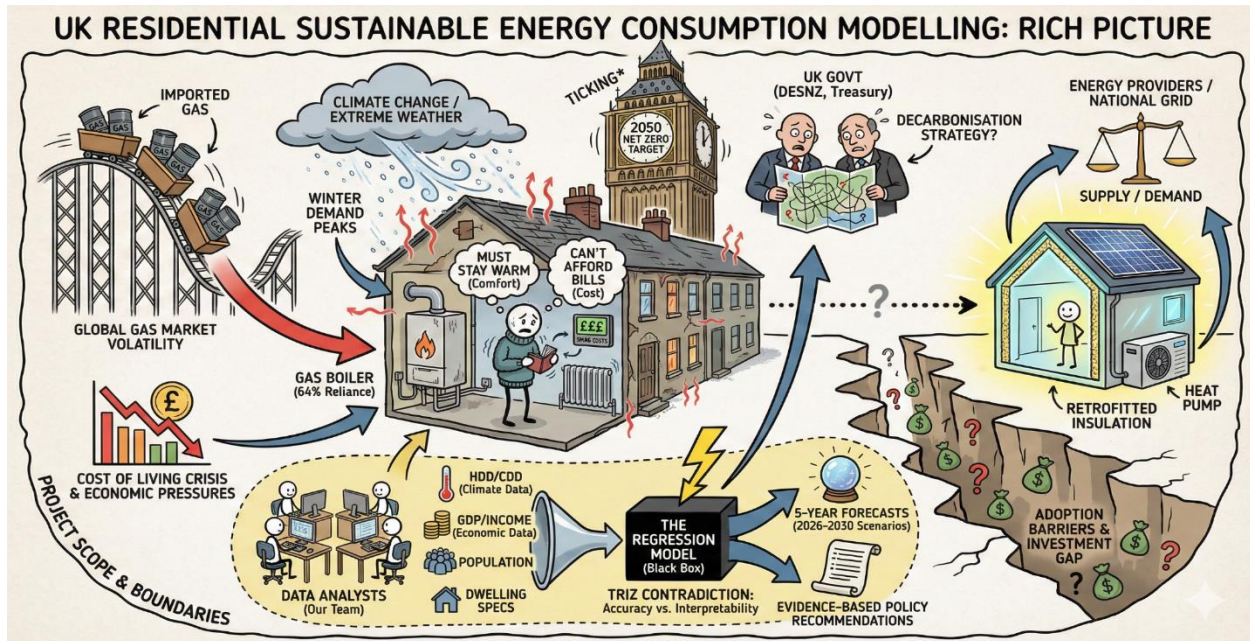
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11. Appendices

11.1 'Rich Picture' Image



Year	total electricity consumption (kWh)	Electricity Price	Gas Price	total natural gas consumption (Gtoe)	Media Disposable Income	Average temperature	Kwh	GVI	Regression	Error	Error%	Logs	EXP	Error	Error%	YTY
1990	8,066	0.71	0.19	25,835	23454	3.58	896600000	807	8,261	195	2%	22.81	8,066	0.00	0%	
1991	8,436	0.84	0.20	26,721	24042	3.54	843600000	844	8,570	134	2%	22.85	8,436	0.00	0%	4%
1992	8,555	0.89	0.20	26,393	24735	3.63	855500000	856	8,579	24	0%	22.71	8,555	0.00	0%	0%
1993	8,639	0.90	0.19	26,254	24032	3.53	863900000	864	8,530	-103	-1%	22.68	8,639	0.00	0%	-1%
1994	8,721	0.92	0.20	26,355	23381	3.83	872000000	872	8,470	-251	-1%	22.83	8,721	0.00	0%	-1%
1995	8,790	0.93	0.21	26,037	24000	3.17	879000000	879	8,500	-210	-2%	22.90	8,790	0.00	0%	1%
1996	9,244	0.90	0.20	32,317	24069	3.18	924396174	924	9,300	-144	-2%	22.85	9,244	0.00	0%	6%
1997	9,382	0.87	0.21	28,710	25403	3.41	89859333	898	9,194	213	2%	22.82	9,382	0.00	0%	1%
1998	9,408	0.80	0.19	30,601	26051	3.16	940800000	941	9,386	-22	0%	22.86	9,408	0.00	0%	2%
1999	9,485	0.77	0.18	30,788	26536	3.37	948500000	949	9,563	83	1%	22.87	9,485	0.00	0%	2%
2000	9,617	0.78	0.19	31,006	26300	3.30	961660000	962	9,878	261	1%	22.89	9,617	0.00	0%	3%
2001	9,917	0.79	0.19	32,625	28776	3.80	991700000	992	10,321	194	1%	23.02	9,917	0.00	0%	0%
2002	10,319	0.75	0.19	32,362	30610	3.44	1031900000	1032	10,366	47	0%	23.06	10,319	0.00	0%	0%
2003	10,576	0.72	0.19	31,232	31600	3.47	1057600000	1058	10,682	106	1%	23.08	10,576	0.00	0%	3%
2004	10,679	0.85	0.22	34,085	32192	3.44	1067900000	1068	10,848	163	2%	23.09	10,679	0.00	0%	2%
2005	10,809	0.83	0.24	32,836	33083	3.42	1080920000	1081	10,871	-138	-1%	23.10	10,809	0.00	0%	-2%
2006	10,723	1.16	0.31	31,550	33465	3.70	1072600000	1072	10,464	-259	-2%	23.10	10,723	0.00	0%	-2%
2007	10,583	1.12	0.36	30,341	33439	3.56	1058600000	1058	10,335	-247	-2%	23.08	10,583	0.00	0%	0%
2008	10,301	1.37	0.41	30,316	33873	3.62	1030100000	1030	10,180	-111	-1%	23.06	10,301	0.00	0%	-1%
2009	10,193	1.36	0.43	29,682	33546	3.14	1019600000	1019	10,033	-163	-2%	23.04	10,193	0.00	0%	-2%
2010	10,218	1.37	0.43	33,439	33591	3.34	1021700000	1022	10,389	182	2%	23.05	10,218	0.00	0%	4%
2011	9,595	1.44	0.51	26,395	33544	3.41	959400000	959	9,629	-35	0%	22.88	9,595	0.00	0%	-1%
2012	9,593	1.54	0.54	25,500	32758	3.74	9593000000	966	9,806	-54	-1%	23.01	9,593	0.00	0%	2%
2013	9,752	1.66	0.56	29,622	32007	3.74	975700000	975	9,682	-53	-1%	22.90	9,752	0.00	0%	-1%
2014	9,293	1.65	0.58	24,393	32830	3.88	9293000000	929	9,172	-						

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Year	total natural gas consumption	Gas Price	Electricity Price	total electricity consumption (Gtoe)	Average temperature	Median Disposable Income	Kwh	GWh	Regression	Error (millions Error 2	Logs	EXP	Error	Error 2	
1930	25,835	0.19	0.77	0,066	9.38	23454	2,584	2584	25,472	363	1.43%	23.374396	25835000000	0.00	0%
1931	28,721	0.20	0.84	0,436	8.54	24042	2,872	2872	28,394	327	1.05%	24.0808344	28721000000	0.00	0%
1932	28,309	0.20	0.83	0,555	8.63	24755	2,839	2839	28,550	-161	-0.56%	24.06326758	28338300000	0.00	0%
1933	28,254	0.19	0.90	8,638	8.33	24032	2,805	2805	28,930	-676	-2.13%	24.03528215	28254000000	0.00	0%
1934	28,355	0.20	0.92	8,721	8.89	23381	2,836	2836	29,465	-1,109	-3.77%	24.06066922	28355000000	0.00	0%
1935	28,037	0.21	0.93	8,790	9.17	24000	2,804	2804	28,366	-329	-3.21%	24.05679303	28037000000	0.00	0%
1936	32,317	0.20	0.90	9,244	8.18	24863	3,232	3232	32,610	-293	-0.90%	24.18884812	32316543422	0.00	0%
1937	23,710	0.21	0.87	8,982	9.41	25403	2,371	2371	29,152	1,058	3.63%	24.11476315	23710404627	0.00	0%
1938	30,601	0.19	0.80	3,408	9.16	26051	3,060	3060	30,543	58	0.19%	24.14423853	30601000000	0.00	0%
1939	30,788	0.18	0.77	3,485	9.37	26598	3,079	3079	30,250	538	1.78%	24.15033084	30788000000	0.00	0%
2000	31,806	0.19	0.78	3,617	9.10	26300	3,181	3181	30,561	1,245	4.07%	24.18234344	31806450000	0.00	0%
2001	32,625	0.19	0.73	3,917	8.80	26776	3,262	3262	32,059	566	1.77%	24.20833735	32624760000	0.00	0%
2002	32,362	0.19	0.75	10,319	9.44	30610	3,236	3236	31,975	387	1.21%	24.20025568	32362160000	0.00	0%
2003	33,232	0.19	0.72	10,576	9.47	31600	3,323	3323	32,522	710	2.18%	24.22677429	33231840000	0.00	0%
2004	34,085	0.22	0.85	10,679	9.44	32192	3,409	3409	32,914	1,171	3.56%	24.25212882	34085190000	0.00	0%
2005	32,836	0.24	0.93	10,609	9.42	33089	3,284	3284	33,155	-319	-0.96%	24.21478167	32836400000	0.00	0%
2006	31,550	0.31	1.16	10,723	9.70	33465	3,155	3155	32,330	-780	-2.41%	24.17484355	31550100000	0.00	0%
2007	30,341	0.36	1.12	10,583	9.56	33439	3,034	3034	31,300	-959	-3.06%	24.1357717	30341180000	0.00	0%
2008	30,316	0.41	1.37	10,301	9.02	33973	3,092	3092	31,257	-941	-3.03%	24.15454163	30316060000	0.00	0%
2009	23,682	0.43	1.36	10,193	9.14	33346	2,968	2968	30,326	-644	-2.13%	24.1137393	23682150000	0.00	0%
2010	33,439	0.43	1.31	10,218	7.94	33591	3,350	3350	32,552	848	2.51%	24.2347858	33439180000	0.00	0%
2011	26,556	0.51	1.44	9,535	9.61	33544	2,656	2656	26,678	-122	-0.46%	24.08250347	2655520000	0.00	0%
2012	29,508	0.54	1.54	9,859	8.74	32758	2,951	2951	29,641	-133	-0.45%	24.10783335	29508180000	0.00	0%
2013	29,624	0.56	1.66	9,752	8.74	32007	2,962	2962	29,614	7	0.03%	24.11177473	29621950000	0.00	0%
2014	24,330	0.58	1.65	9,283	9.88	32830	2,438	2438	25,190	-757	-3.05%	23.91756327	24330200000	0.00	0%
2015	25,587	0.57	1.67	9,266	9.18	34255	2,553	2553	26,000	-413	-1.53%	23.96536666	25587420000	0.00	0%
2016	26,000	0.56	1.71	9,288	9.29	34954	2,600	2600	25,938	62	0.24%	23.98104584	2599570000	0.00	0%
2017	25,432	0.52	1.80	9,056	9.53	35794	2,543	2543	24,990	442	1.77%	23.95521117	25431940000	0.00	0%
2018	26,045	0.53	1.95	9,111	9.45	34308	2,604	2604	26,084	-39	-0.07%	23.98305819	26044190000	0.00	0%
2019	25,144	0.50	2.02	8,941	9.39	35224	2,514	2514	25,915	-771	-3.07%	23.94789306	25144350000	0.00	0%
2020	25,813	0.46	2.03	9,250	9.62	36671	2,581	2581	26,785	-972	-3.63%	23.97414524	25813030000	0.00	0%
2021	27,432	0.48	2.25	9,166	9.38	37414	2,741	2741	27,287	124	0.46%	24.03423029	2741550000	0.00	0%
2022	22,322	0.30	3.27	8,259	10.03	37006	2,232	2232	21,467	854	3.98%	23.82882046	22321600000	0.00	0%
2023	20,880	1.10	3.76	7,968	9.97	36394	2,089	2089	20,257	632	3.12%	23.7624579	20883600000	0.00	0%
2024	21,754	0.86	3.24	8,115	9.79	36663	2,175	2175	21,846	-92	-0.42%	23.80304634	21753640000	0.00	0%
2025	21,445	0.88	3.30	7,842	9.77	37095.571	2,059	2059	20,595	850	4.1%				
2026	21,253	0.90	3.57	7,650	9.80	37328.142	2,035	2035	20,341	906	4.5%				
2027	21,062	0.92	3.83	7,458	9.82	37960.713	2,010	2010	20,099	962	4.8%				
2028	20,870	0.94	4.10	7,265	9.84	38393.285	1,985	1985	19,852	1,018	5.1%				
2029	20,679	0.96	4.36	7,073	9.86	38825.856	1,960	1960	19,604	1,075	5.5%				
Electricity Price															
Scenario 1 Scenario 2															
-15%	3.30	2.80													
-33%	3.37	2.40													
-45%	3.83	2.00													
-54%	4.10	1.90													
-58%	4.36	1.85													

SUMMARY OUTPUT

Regression Statistics									
Multiple R	0.981500749								
R Square	0.96334372								
Adjusted R Square	0.957023672								
Standard Error	717922705.5								
Observations	35								
ANOVA									
	df	SS	MS	F	Significance F				
Regression	5	3.92813E+20	7.85627E+19	152.4266404	6.75413E-20				
Residual	29	1.4947E+19	5.15413E+17						
Total	34	4.0776E+20							
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%	
Intercept	16723502628	3321293005	5.035238565	0.00	9930695723	23516309533	9930695723	23516309533	
Gas Price	-1.07E+10	2056598870	-5.202672388	0.00	-14906027127	-6493593185	-14906027127	-6493593185	
Electricity Price	3592846548	815391198.7	4.406285662	0.00	1925184299	5260508798	1925184299	5260508798	
total electricity consumption (Gtoe)	4.11654641	0.354074748	11.62620727	0.00	3.392382239	4.84071058	3.392382239	4.84071058	
Median Disposable Income	-362747.974	78619.12864	-4.61399128	0.00	-523542.1464	-201953.8017	-523542.1464	-201953.8017	
Average temperature	-1781094921	303150343.3	-5.875285845	0.00	-2401106989	-1161082853	-2401106989	-1161082853	